

Every term at $d = 3$:

$Z_3(n) = n^{-1/2} P(\log n) + O(n^{-1})$ with P an explicit quadratic

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Abstract

Unit 38 gave the leading term $Z_3(n) \sim \frac{A}{8} n^{-1/2} (\log n)^2$ for the $d = 3$ chart integral with three coinciding exponents. Iterating the tail mechanism of unit 41 once more identifies the whole polynomial: with $\theta(x) = H(x) - A \log x - C_0 \in [0, M/x]$ for $x \geq 1$,

$$H_2(L) = \frac{A}{2} \log^2 L + C_0 \log L + C_1 - \int_L^\infty \frac{\theta(x)}{x} dx, \quad 0 \leq \int_L^\infty \frac{\theta}{x} \leq \frac{M}{L},$$

with $C_1 = \int_0^1 H/x + \int_1^\infty \theta/x$, and hence $Z_3(n) = n^{-1/2} P(\log n) + O(n^{-1})$ where $P(\ell) = \frac{A}{8} \ell^2 + (\frac{3}{2} A \log b + \frac{1}{2} C_0) \ell + (\frac{9}{2} A \log^2 b + 3 C_0 \log b + C_1)$. Zero sorry/axiom.

1 Setting

The recursion $H_{m+1}(L) = \int_0^L H_m(x)/x dx$ maps an expansion “polynomial in $\log x$ plus a remainder $O(1/x)$ ” to an expansion of the same shape one degree higher: the polynomial part integrates exactly against dx/x over $[1, L]$, the remainder over $[1, \infty)$ converges to a new constant, and its tail over $[L, \infty)$ is again $O(1/L)$. This unit runs the step once, from H (unit 41) to H_2 .

2 The things formalised

```

noncomputable def logRemainder ( a x : ) : :=
  logPrimitive a x - quadMass a * Real.log x - logConst a
theorem logRemainder_div_integrableOn ... (hc : 1 < c) :
  IntegrableOn (fun x => logRemainder a x / x) (Ioi c)
noncomputable def logConst2 ( a : ) : :=
  ( x in Ioc 0 1, logPrimitive a x / x) + x in Ioi 1, logRemainder a x / x
theorem logLogPrimitive_sub_poly ... (hL : 1 < L) :
  logLogPrimitive a L
  - (quadMass a * Real.log L ^ 2 / 2 + logConst a * Real.log L + logConst2 a)
  = - x in Ioi L, logRemainder a x / x
theorem threeDIntegral_second_order_isBig0 ... :
  (fun n => threeDIntegral a b n - n ^ (-(1/2 : )) * threeDSecondOrderCoeff a b n)
  = 0[atTop] fun n => n ^ 1
theorem threeDIntegral_second_order_tendsto ... :
  Tendsto (fun n => Real.sqrt n * threeDIntegral a b n
    - quadMass a / 8 * Real.log n ^ 2
    - (3 * quadMass a * Real.log b / 2 + logConst a / 2) * Real.log n)
  atTop ( (9 * quadMass a * Real.log b ^ 2 / 2 + 3 * logConst a * Real.log b
    + logConst2 a))

```

3 Proof strategy

On $(1, L]$ the integrand $H(x)/x$ is rewritten pointwise as $A \log x/x + C_0 x^{-1} + \theta(x)/x$; the first two integrate to $\frac{A}{2} \log^2 L + C_0 \log L$ by the FTC lemmas of unit 38, and the third is $\int_1^\infty \theta/x - \int_L^\infty \theta/x$ by `setIntegral_union`. Combined with the split at $x = 1$ this is the exact identity. The integrability of $\theta(x)/x$ on $[1, \infty)$ uses the domination $0 \leq \theta/x \leq Mx^{-2}$ from unit 41 and `integrableOn_Ioi_rpow_of_lt`; measurability comes from continuity on $(1, \infty)$, where H agrees with its interval-integral primitive and $\log$ is continuous. The Z_3 statements then follow from $Z_3 = n^{-1/2} H_2(\sqrt{n}b^3)$ and $\log(\sqrt{n}b^3) = \frac{1}{2} \log n + 3 \log b$ exactly as at $d = 2$; the polynomial coefficient is kept in nested form and expanded by a separate `ring` lemma so the `Tendsto` statement can name the constant. The identity and remainder sign were checked numerically first (residual 0 at $L \geq 5$ to double precision).

4 Roadblocks and resolutions

- `Integrable.add` returns a `Pi.add`; `rw [integral_add ...]` then fails to match the λ -integrand. Type-ascribe the combined integrability witness as a single lambda (the documented workaround) before rewriting.
- `positivity` cannot see $0 \leq M$ for the opaque `quadMoment`; supply `quadMoment_nonneg` through `mul_nonneg/div_nonneg`.
- `field_simp` on the pointwise decomposition of $H(x)/x$ leaves a commutativity goal; follow with `ring`.

5 What was Mathlib, what was new

Mathlib: `ContinuousOn.aestronglyMeasurable`, `integrableOn_Ioi_rpow_of_lt`, `integral_Ioi_rpow_of_lt`, `setIntegral_union`, `squeeze_zero`'. New: the remainder θ and constant C_1 of H_2 , and the complete $O(n^{-1})$ -accurate expansion of the $d = 3$ log-squared chart integral.

6 Lessons

The one-step transfer “polynomial in $\log$ plus $O(1/x)$ ” $\mapsto$ same shape one degree higher is now visible as a reusable pattern; a general- m version for `iterLogPrim` would make the constants below $(\log n)^{d-1}$ available for every d at once, mirroring how unit 39 generalised units 35 and 38.

7 Outcome

The $d = 2$ and $d = 3$ coinciding-exponent integrals now carry complete polynomial-in- $\log n$ expansions with explicit constants and $O(n^{-1})$ remainders. Next candidates: the general- m transfer lemma for `iterLogPrim`, or the weighted primitives F_γ for general (k, h) .